Supplementary Information

Modeling Protein Synthesis

by

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1 | Methods

The following section discusses the methods used in the report of which utilize VODE solvers in the *ddeint package for Python. These methods were pioneered by Brown in 1989 [1].*

1.0.1 Adams-Bashforth

Adams methods are based on the idea of approximating the integrand with a polynomial within the interval (t_n, t_{n+1}). Using a kth order polynomial results in a $k+1$th order method. There are two types of Adams methods, the explicit and the implicit types. The explicit type is called the Adams-Bashforth (AB) methods and the implicit type is called the Adams-Moulton (AM) methods [2], which is used when the ODE is not stiff.

$$y(t_{n+1}) = y(t_n) + \int_{t_n}^{t_{n+1}} f(t, y(t))dt \tag{1.1}$$

1.0.2 Backward Differentiation Approach

The VODE solver falls back to the backward differentiation approach whenever the system is stiff. The general formula used is as follows [3].

$$y_n = y_{n-1} + (t_n - t_{n-1})f(y_n, t_n) \tag{1.2}$$

1.0.3 Parameters

Parameter	Value	Description
c	200	Basal mRNA production rate
a	30	Repression threshold
b	0.9	Repression sensitivity
k	0.6	mRNA degradation rate
e	0.4	Translation rate
f	0.4	Enzyme degradation rate
g	0.2	Repressor production rate
h	0.2	Repressor degradation rate

Table 1.1: Model parameters used in all simulations, chosen to produce behavior consistent with Maynard Smith.

1.1 Goodwin oscillator

The Goodwin oscillator takes the form of the following system of ODEs where $k_1, k_2, \cdots, k_6$ are parameters [4].

$$\frac{dX}{dt} = k_1 \frac{K_i^n}{K_i^n + Z^n} - k_2 X \tag{1}$$

$$\frac{dY}{dt} = k_3 X - k_4 Y \tag{2}$$

$$\frac{dZ}{dt} = k_5 Y - k_6 Z \tag{3}$$

Some of the results from the paper cited above show extremely similar results to those produced in the main report.

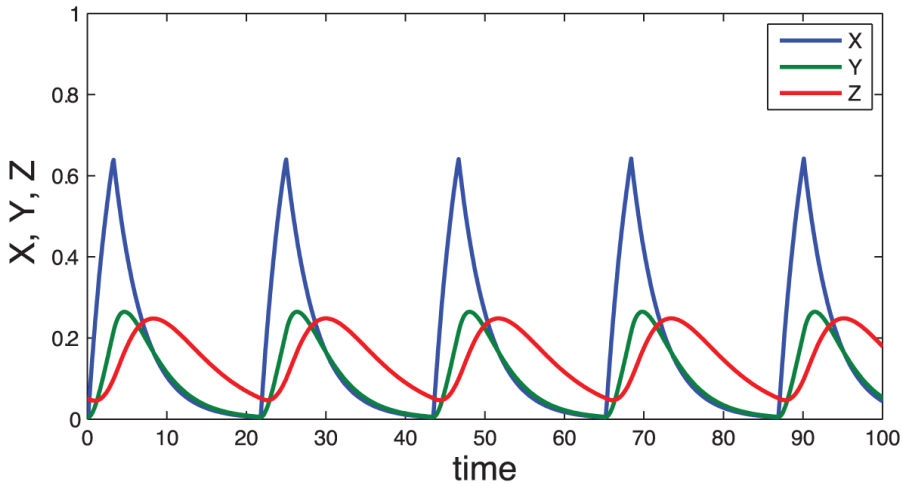

Figure 1.1: Simulation of Goodwin oscillator model showing X, Y and Z amplitudes over 100 time units

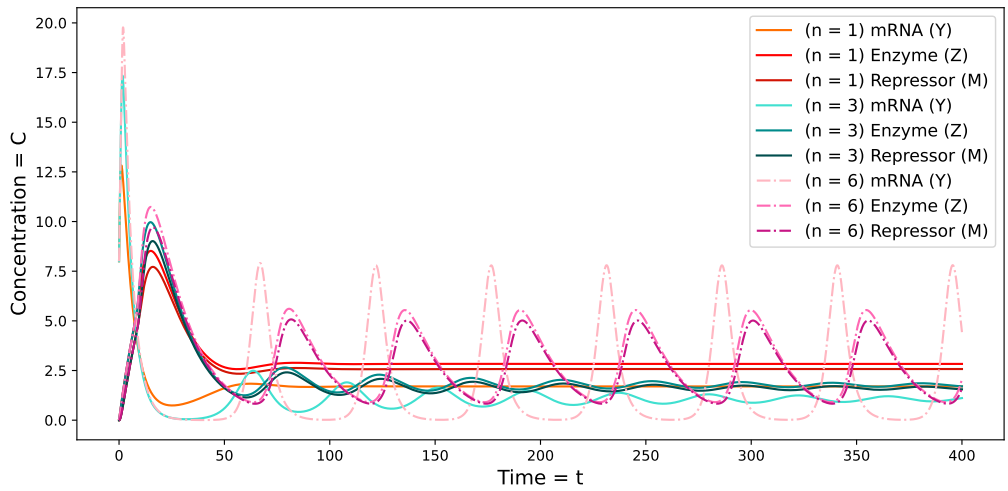

Figure 1.2: Simulation of delayed model ($T = 8$) incorporating coefficient changes ($n = 1$) vs. ($n = 3$) vs. ($n = 6$) showing mRNA, enzyme and repressor molecule concentrations over 400 time units evaluated at 4000 intervals.

Bibliography

[1] P. N. Brown, G. D. Byrne, and A. C. Hindmarsh, "VODE: A Variable-Coefficient ODE Solver," SIAM Journal on Scientific and Statistical Computing, vol. 10, no. 5, pp. 1038–1051, 1989. Note: The original paper describing the VODE algorithm's unique variable-coefficient approach.

[2] M. Zeltkevic, "Adams Methods," MIT Course 10.001: Introduction to Computer Methods, Course Notes on Differential Equations, 1998. Available at: https://web.mit.edu/10.001/Web/Course_Notes/Differential_Equations_Notes/node6.html. Note: Provides a foundational overview of explicit Adams-Bashforth and implicit Adams-Moulton methods used in multi-step solvers.

[3] J. D. Lambert, Numerical Methods for Ordinary Differential Systems: The Initial Value Problem, John Wiley & Sons: Chichester, 1991. Note: Provides thorough analysis of stability regions (A-stability) for multistep methods.

[4] D. Gonze and W. Abou-Jaoudé, "The Goodwin Model: Behind the Hill Function," PLOS ONE, vol. 8, no. 8, p. e69573, 2013. doi: 10.1371/journal.pone.0069573. Note: Investigation of the biochemical origins of the Hill coefficient and its role in the stability of the Goodwin oscillator.